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Project Synopsis (2025-26) - Sem VII

INDREVA: India Real Estate Valuation and Risk Assessment Framework

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Abstract

India's push for infrastructure and housing is accelerating, with the Union Budget (FY 2024–25) allocating **₹11.11 lakh crore** [1] towards **capital expenditure** and **promoting market-based financing**. However, the real estate sector continues to face challenges in **accurate property valuation, transparency, data fragmentation** as practices remain subjective and often rely on inconsistent, locally updated guideline values. Despite recent reforms and standardisation efforts by regulators and professional bodies, these gaps lead to **market distortions, mispricing,** and **higher financial risks** for **buyers, investors, and lenders**. This project aims to develop a real estate valuation & risk assessment framework and platform that integrates diverse public & proprietary data sources and by using Big Data Analysis, Machine Learning techniques & AI offer **standardised property valuations, risk scores,** and **market insights** enabling stakeholders to make data-backed decisions. It intends to democratize access to critical property information, thus enhancing **transparency, reducing mispricing,** and **supporting sustainable urban growth** within the Indian real estate ecosystem.

Introduction

Recent studies and reports continue to project India's urban population to reach **about 600 million by 2036** [2]. This projection is being used by planners and think-tanks to argue for major scaling of urban infrastructure, climate adaptation and financing. India's real estate sector is entering a transformative phase driven by rapid urbanisation, institutional investment, and policy reforms. With Tier-1 cities entering the "transparent" category in **JLL's Global Real Estate Transparency Index 2024**, India has become the top global improver—thanks to digitised land records, streamlined building regulations, and climate-risk disclosure mandates [3] [4]. At the same time, major developers in the sector now collectively hold a valuation of roughly **₹16 lakh crore—surpassing Kuwait's GDP**—highlighting the immense economic significance of the market.

Despite this progress, the sector continues to grapple with persistent structural challenges. Real estate projects frequently face delays or lapse altogether—as evidenced in Karnataka, where more than **2,600 projects have expired** and another **2,700 have lapsed**—undermining regulatory confidence and buyer trust [5]. High transaction costs, opaque processes, and legacy legal bottlenecks also contribute to mispricing, informal payments, and the use of black money in property deals [6]. Public policy initiatives like RERA (enacted in 2016) have improved dispute resolution and mandatory registration, but enforcement remains inconsistent across states [7]. Meanwhile, innovation in PropTech, AI, blockchain tokenisation, and smarter analytics holds promise for enhancing transparency, unlocking liquidity, and improving valuation accuracy—but adoption remains fragmented [8].

Against this backdrop, there is a need for a **unified, data-driven platform** that merges public records, registries, RERA data, transaction history, GIS layers, and economic indicators—augmented by AI and ML techniques, BDA & AI—to deliver **standardised property valuations**, **risk assessments**, and **market insights**. Such a framework would not only democratise access to critical property information but also support better decision-making, reduce financial risks, and promote sustainable urban growth in India's evolving real estate ecosystem.

Problem Statement

India's real estate market, despite its rapid growth, lacks a **standardized, transparent, and dynamic valuation** framework. **Fragmented** and **outdated data sources**, **inconsistent valuation methods**, and **opaque pricing** mechanisms lead to uninformed buying decisions for **homebuyers**, increased credit risk for **banks**, poorly aligned project planning for **developers**, reduced **investor** confidence, and weak infrastructure forecasting for **urban planners**. This lack of clarity not only raises the risk of financial losses and non-performing assets but also fuels speculative practices, slows sustainable market growth, and limits effective policy-making. Addressing this challenge requires a **unified, data-driven platform** capable of integrating diverse datasets—from property listings, government registries, RERA records, and macroeconomic indicators to geospatial risk data—and applying advanced analytics. This project will develop a framework using BDA, ML & AI to deliver **standardized property valuations**, comprehensive **risk assessments**, and **actionable market insights**, empowering all stakeholders to make informed, evidence-based decisions.

Proposed Solution

This project proposes the development of the following components:

Data Aggregation

- **Property Transaction Data:** Scraping and consolidation of property transaction data from key state registration portals like Maharashtra's e-Registration (IGR e-filing), Karnataka Kaveri, and Delhi DORIS—to capture actual sale records and legal transaction details.
- **Regulatory & Project Data:** Aggregation of RERA-approved project information from all state RERA websites, including project status, developer profiles, and legal histories.
- **Satellite & GIS Data:** Integration of satellite imagery and spatial data from ISRO's Bhuvan platform or Google Earth Engine to analyze urban growth, infrastructure development, and environmental risks such as flood-prone or seismic zones.
- **Government & Macroeconomic Data:** Inclusion of demographic data from Census 2011 and 2021 updates, macroeconomic indicators such as GDP, inflation, and RBI repo rates to contextualize market dynamics.

Valuation and Risk Calculation

- **Machine Learning-Based Price Prediction:** Use machine learning algorithms and AI trained on consolidated datasets for accurate property price forecasting. Models will incorporate geospatial features, historic transactions, and macroeconomic signals to improve prediction accuracy.
- **Multi-Factor Risk Scoring:** Development of a risk scoring system incorporating:
    - **Natural Disaster Risk:** Data from the National Disaster Management Authority (NDMA) on floods, cyclones, earthquakes.
    - **Socioeconomic Risk:** Crime rates, unemployment statistics, and industrial activity indices sourced from the National Crime Records Bureau (NCRB) and Ministry of Labour & Employment.
    - **Regulatory & Developer Risk:** Developer track record evaluation based on RERA compliance history and litigation frequency.

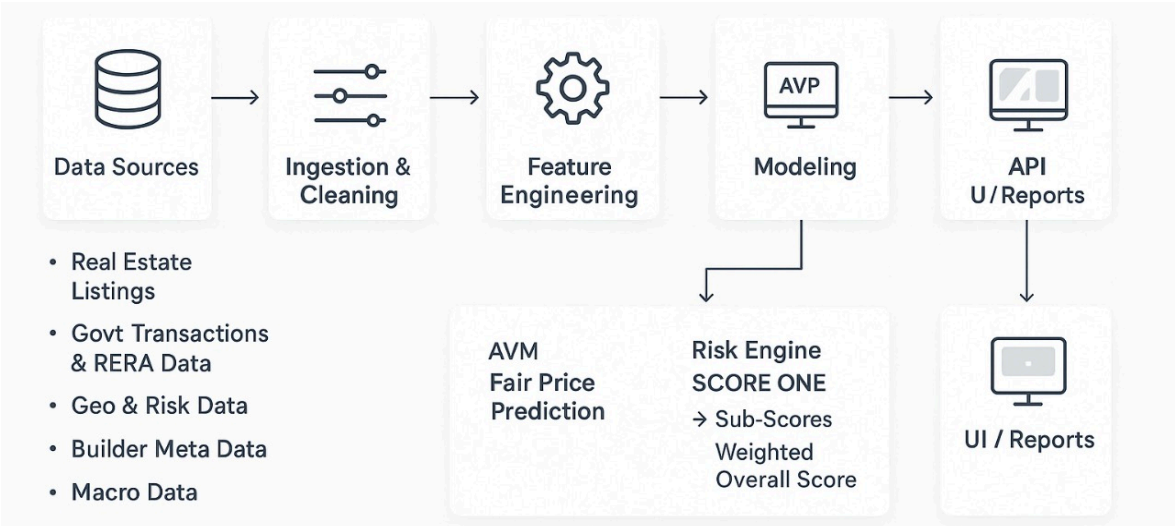
User-Centric & Public-Facing Interface

- **Automated Dashboards:** Dashboards showing property appreciation trends, distressed asset clusters, and infrastructure impacts.
- **Public Utility Tool:** Tool to check fair market values and risk scores for any property, enhancing transparency.
- **API Services:** Secure, RESTful APIs for banks, NBFCs, developers, and fintech startups to integrate valuation and risk data into their own applications.

Methodology / Block Diagram

The project will follow a phased, modular development approach:

1. **Data Acquisition and Pre-processing:**
    - Develop automated scrapers and API connectors for data ingestion from state portals, RERA sites, government open data platforms, and satellite imagery providers.
    - Cleanse, normalize, and unify heterogeneous datasets into a consistent schema.
    - Perform exploratory data analysis to identify key features and patterns.
2. **Model Development:**
    - Build, train, and validate multiple ML models for price prediction, including ensemble methods and neural networks.
    - Engineer composite risk scoring algorithms by weighting various risk factors.
    - Validate models with historical data and expert feedback.
3. **Platform & Interface Development:**
    - Backend development using robust frameworks like Django or Flask with PostgreSQL and MongoDB for geospatial and unstructured data.
    - Frontend development using React or Vue.js to build user-friendly dashboards and tools.
    - Design and deploy RESTful APIs for external integration.



Hardware, Software and Tools Requirements

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel i5 8th Gen / AMD Ryzen 5	Intel i7 / AMD Ryzen 7
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
GPU (Optional)	Integrated GPU	NVIDIA GTX 1650 or better / Cloud GPU (Google Colab)
Internet	5 Mbps broadband	20+ Mbps broadband

Software Requirements

Category	Tool / Library	Version / Notes	Purpose
Operating System	Windows 10/11, Ubuntu 22.04 LTS, macOS 13+	Latest stable	Development environment
Programming Language	Python	3.10+	Core programming language
	SQL	Any standard version	Database queries
Data Handling & BDA	Pandas	Latest stable	Data manipulation and analysis
	Dask	Latest stable	Scalable parallel data processing

	PySpark (local mode)	Latest stable	Big data processing in standalone mode
Machine Learning & GNN	PyTorch	Latest stable	Deep learning and graph neural networks
	PyTorch Geometric	Latest stable	Graph neural network implementation
	Scikit-learn	Latest stable	Traditional machine learning algorithms
Geo-Spatial Analysis	GeoPandas	Latest stable	Geospatial data processing
	Folium	Latest stable	Interactive map visualizations
	Shapely	Latest stable	Geometric operations on spatial data
Data Visualization	Matplotlib, Seaborn, Plotly	Latest stable	Charts and interactive visualization
Data Cleaning	OpenRefine (optional)	Latest stable	GUI-based data cleaning
Web Framework	Django	Latest stable	Backend API development
Dashboard Framework	Streamlit or Plotly Dash	Latest stable	Interactive dashboards with minimal coding
UI Styling	Bootstrap or Tailwind CSS	Latest stable	Frontend styling
Databases	PostgreSQL + PostGIS	Latest stable	Relational + spatial queries
	SQLite (for prototyping)	Latest stable	Lightweight local DB

Tools and Platforms

Tool / Platform	Purpose
VS Code	Code editing and IDE
GitHub	Version control and collaboration
Google Colab	Free cloud GPU for training ML/GNN models
QGIS (optional)	GUI-based geospatial analysis
Kaggle Datasets	Public datasets for real estate, finance, geography
Mendeley / Zotero	Research paper management and referencing

Proposed Evaluation Measures

1. Model Accuracy & Performance :

- **Price Prediction Accuracy:**
  - **Metrics:** Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and R^2 score.
  - **Target:** MAPE ≤ 10% for metro cities and ≤ 15% for Tier-2/3 cities.
- **Risk Scoring Validation:**
  - Cross-validation with historical loss data from banks/NBFCs.
  - Expert review by real estate valuers and risk analysts.

2. Data Coverage & Completeness :

- Percentage of properties covered in major metro and tier-2/3 cities (target ≥ 85% in pilot regions).
- Completeness score for each property dataset — property transaction details, regulatory history, geospatial risk factors.

3. System Performance :

- **API Latency:** Average response time ≤ 500ms for valuation requests.
- **Data Freshness:** New transaction and regulatory data integrated within ≤ 48 hours of availability.

4. Usability & User Experience :

- User feedback surveys using the System Usability Scale (SUS) — target SUS score ≥ 80.
- A/B testing for public dashboards and API documentation clarity.

5. Scalability & Robustness :

- System load testing for up to 10,000 concurrent valuation requests.
- Failover mechanisms in case of data source unavailability.

Conclusion

This project addresses the need for a standardized, transparent, and data-driven real estate valuation and risk assessment framework in India. By integrating diverse datasets—including transaction records, regulatory information, geospatial data, and macroeconomic indicators—and leveraging advanced machine learning and graph-based models, the platform aims to provide accurate property valuations and comprehensive risk scores. The proposed solution will empower buyers, investors, lenders, and policymakers with actionable insights, improving market transparency and reducing financial risks. Ultimately, this initiative supports sustainable urban growth and enhances trust within India's evolving real estate ecosystem, making it a valuable contribution to the sector's digital transformation.

References

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